

ADVANCED ENGINEERING & STATISTICAL ANALYTICS

Comprehensive Reference Manual: Chapters 5 & 6 Extended

Theoretical Axioms, Detailed Derivations, and High-Density Multi-Domain Practical Problem Sets

VOLUME MASTER INDEX & TECHNICAL MAP

This master compilation explicitly merges theoretical proofs, structural operational conditions, and complex empirical testing scenarios across industrial, economic, software architectural, and biological domains.

Module Theme	Core Analytical Topics Covered
Module 1: Sampling Concepts	Populations, Frameworks, Random vs Non-random distributions, Bias corrections, Degrees of Freedom.
Module 2: Limit Theorems	Central Limit Theorem mechanics, Finite Population Correction (FPC), Standard Error structures.
Module 3: Estimation Theory	Point vs Interval estimation, MVUE, Confidence bounds for means and proportions.
Module 4: Decision Dynamics	Hypothesis frameworks, Type I and II error paths, Power analysis, Z-tests vs t-tests.
Module 5: Categorical Tests	Chi-Square distribution, Goodness-of-fit mechanics, Contingency tables, Independence testing.
Module 6: Variance Partitioning	ANOVA models, Sum of Squares identities, Mean Squares, F-distribution validations.
Module 7: Process Systems	Statistical Quality Control, Control charts (X-bar, R, p, c), 3-Sigma limits, Process stability.

1. Extended Foundational Concepts of Sampling

1.1 Structural Taxonomy of Data Domains

To mathematically analyze populations, we must understand how individual points of data are grouped together. A **Population** comprises the complete set of all possible observations that exist within a defined specification. In analytical environments, this set is defined by mathematical parameters that are usually fixed but unknown. Conversely, a **Sample** represents a calculated slice of reality, a finite sequence of observations drawn systematically or randomly from the population domain.

The distinction between parameters and statistics is fundamental to tracking errors:

- **Population Parameters:** Calculated by gathering information from the entire target universe. They are completely static and do not change unless the underlying population changes. Symbolized by Greek notations: Mean (μ), Variance (σ^2), Standard Deviation (σ), and Proportion (P).
- **Sample Statistics:** Calculated exclusively from observed sample sequences. These change with each new sample draw, making them random variables. Symbolized by Roman notations: Mean ($\bar{x}$), Variance (s^2), Standard Deviation (s), and Proportion (p).

1.2 Random Sampling Frameworks

To perform valid statistical calculations, samples must be selected without personal bias. The most common methods are detailed below:

1.2.1 Simple Random Sampling (SRS) Mechanics

In simple random sampling, every single item in the population has an equal chance of being selected during each selection step. It is divided into two distinct mathematical models:

1. **With Replacement (SRSWR):** After an item is selected from the population, it is recorded and then returned to the pool before the next draw. Because of this, the population size stays exactly the same (N) for every single draw, and individual items can be selected multiple times. The total number of unique samples of size n that can be drawn from a population of size N is equal to:

$$\text{Total Samples (SRSWR)} = N^n$$

2. **Without Replacement (SRSWOR):** After an item is selected, it is removed from the population pool, so it cannot be selected again. This causes the remaining population to shrink by one item at each

successive stage of the sample selection. The total number of unique samples of size n that can be extracted from a finite population of size N is determined by the combination formula:

$$\text{Total Combinations (SRSWOR)} = N! / (n! * (N - n)!)$$

1.2.2 Stratified Random Sampling and Optimization Models

When a population is heterogeneous but contains distinct subgroups, stratified random sampling significantly improves estimation accuracy. The population is divided into non-overlapping groups called strata, where each stratum is internally homogeneous. Simple random samples are then drawn independently from each individual stratum.

Researchers allocate sample sizes to each stratum using two primary methods:

- **Proportional Allocation:** The sample size of each stratum is directly proportional to its total size within the overall population.

$$n_i = n * (N_i / N)$$

- **Optimum Allocation (Neyman Allocation):** This method balances both the size of the stratum and its internal variance, focusing more sampling effort where variability is highest to minimize overall estimation error.

$$n_i = n * (N_i * \sigma_i) / [\sum (N_j * \sigma_j)]$$

1.2.3 Systematic Selection Protocols

Systematic sampling offers an efficient alternative when data is organized sequentially. The selection process calculates a fixed sampling interval (k) based on the population size and the desired sample size:

$$k = N / n$$

A random starting number (r) is selected between 1 and k . The final sample then consists of the items corresponding to the sequence: $r, r + k, r + 2k, r + 3k, \dots, r + (n - 1)k$.

2. Advanced Analysis of the Central Limit Theorem and Standard Errors

2.1 Core Theoretical Principles

The Central Limit Theorem (CLT) provides the mathematical justification for using normal distributions in inferential statistics. It guarantees that as long as the sample size is sufficiently large, the distribution of sample means will follow a normal curve, regardless of the underlying shape of the population distribution.

The primary metrics of the sampling distribution are defined below:

- **Expected Mean:** The average of all possible sample means is exactly equal to the population mean.

$$E(\bar{x}) = \mu$$

- **Standard Error of the Mean (Infinite Populations or SRSWR):** Measures the standard deviation of the sample means, representing how much they vary around the true population mean.

$$SE(\bar{x}) = \sigma / \sqrt{n}$$

- **Standard Error with Finite Population Correction (SRSSWOR):** When sampling without replacement from a finite population, the standard error shrinks as the sample captures a larger portion of the population. We apply the Finite Population Correction (FPC) factor whenever the sample size exceeds 5% of the total population ($n / N > 0.05$):

$$SE(\bar{x})_{\text{finite}} = (\sigma / \sqrt{n}) * \sqrt{ (N - n) / (N - 1) }$$

2.2 Properties of Sample Proportions

When analyzing categorical attributes (such as pass/fail conditions), we evaluate the sample success proportion ($p = x / n$). When the sample size is large enough ($n * P \geq 5$ and $n * (1 - P) \geq 5$), the sampling distribution of proportions approaches a normal distribution, featuring the following parameters:

- **Expected Value:** $E(p) = P$

- **Standard Error of Proportions:**

$$SE(p) = \sqrt{ P * (1 - P) / n }$$

3. Comprehensive Theory of Point and Interval Estimation

3.1 Evaluating Estimator Efficiency

An estimator uses sample data to approximate an unknown population parameter. To determine if an estimator is reliable, it must meet four critical criteria:

1. **Unbiasedness:** An estimator is unbiased if its average value across all possible samples equals the true parameter. For instance, the sample mean is unbiased because $E(\bar{x}) = \mu$. However, using a divisor of n to calculate sample variance underestimates the true population variance. To remove this bias, we must use a divisor of $n - 1$:

$$s^2 = \sum (x_i - \bar{x})^2 / (n - 1)$$

2. **Consistency:** An estimator is consistent if its value converges closer to the true population parameter as the sample size increases toward infinity, causing its standard error to approach zero.
3. **Efficiency:** When comparing two unbiased estimators, the one with the smaller variance across samples is more efficient, as it provides more precise estimates.
4. **Sufficiency:** An estimator is sufficient if it utilizes all the relevant information contained within the sample data to estimate the target parameter.

3.2 Structural Confidence Boundaries

Interval estimation accounts for sampling uncertainty by creating a range of values where the true population parameter is expected to fall. The interval is constructed around a chosen confidence level, denoted as $1 - \alpha$.

$$\text{Confidence Limits} = \text{Point Estimate} \pm (\text{Critical Reliability Factor} \times \text{Standard Error})$$

3.2.1 Confidence Limits for a Single Mean

- **Large Sample Framework ($n \geq 30$) or Known Population Variance:** Uses the standard normal Z-distribution. If the population standard deviation σ is unknown, the sample standard deviation s is substituted:

$$\text{Limits} = \bar{x} \pm Z_{\alpha/2} * (s / \sqrt{n})$$

- **Small Sample Framework ($n < 30$) and Unknown Variance:** If the underlying population is normally distributed, we use the Student's t-distribution with **$df = n - 1$** degrees of freedom:

$$\text{Limits} = \bar{x} \pm t_{\alpha/2, n-1} * (s / \sqrt{n})$$

3.2.2 Confidence Limits for a Single Proportion

The confidence boundaries for an unknown population proportion **P** are calculated using the sample proportion **p** and sample size **n** :

$$\text{Limits} = p \pm Z_{\alpha/2} * \sqrt{(p * (1 - p) / n)}$$

3.2.3 Confidence Limits for the Difference of Two Independent Means

- **Large Independent Samples:** Compares means between two distinct groups using large samples:

$$\text{Limits} = (\bar{x}_1 - \bar{x}_2) \pm Z_{\alpha/2} * \sqrt{(s_1^2 / n_1 + s_2^2 / n_2)}$$

- **Small Independent Samples with Equal Variances:** When group variances are assumed equal, we combine them into a pooled sample standard deviation (**s_p**) and use the t-distribution with **$df = n_1 + n_2 - 2$** :

$$s_p = \sqrt{[(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2] / (n_1 + n_2 - 2)}$$

$$\text{Limits} = (\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2, n_1+n_2-2} * s_p * \sqrt{(1/n_1 + 1/n_2)}$$

3.2.4 Confidence Limits for the Difference of Two Independent Proportions

To compare success rates across two independent populations, use the following large-sample formula:

$$\text{Limits} = (p_1 - p_2) \pm Z_{\alpha/2} * \sqrt{(p_1 * (1 - p_1) / n_1) + (p_2 * (1 - p_2) / n_2)}$$

4. Hypothesis Testing Methodology & Error Control

4.1 Theoretical Foundations of Decision Criteria

Hypothesis testing is a structured statistical framework used to evaluate claims about populations based on sample evidence. It balances two competing statements:

- **Null Hypothesis (H_0):** The baseline assumption that there is no effect, no change, or no difference between groups. It establishes an equality statement that is assumed true unless the sample evidence strongly contradicts it.
- **Alternative Hypothesis (H_1):** The claim the researcher wants to support, asserting a significant change, effect, or difference. It can be directional (one-tailed) or non-directional (two-tailed).

The significance level (α) defines the maximum acceptable risk of committing a Type I error. The chosen value of α establishes the size of the rejection region along the distribution curve. If the calculated test statistic falls within this region, we reject the null hypothesis because the observed data would be highly improbable under the baseline assumption.

4.2 Managing Statistical Power and Errors

Every hypothesis test carries a risk of two distinct errors, which exist in an inverse relationship:

1. **Type I Error (α):** Occurs when we reject a null hypothesis that is actually true (a false positive). This risk is controlled directly by setting the significance level.
2. **Type II Error (β):** Occurs when we fail to reject a null hypothesis that is actually false (a false negative). The probability of a Type II error depends on factors like sample size, population variability, and the actual size of the effect.
3. **Statistical Power ($1 - \beta$):** The probability of correctly rejecting a false null hypothesis. To increase power, researchers can increase the sample size, raise the significance level, or reduce measurement error.

5. Chi-Square Testing Frameworks for Categorical Data

5.1 Properties of the Chi-Square Distribution

The Chi-Square (χ^2) distribution is a continuous probability distribution that is skewed to the right and bounded below by zero. Its specific shape is determined entirely by its degrees of freedom (**df**). As the degrees of freedom increase, the distribution becomes more symmetrical, eventually approaching a normal curve.

The standard formula used to evaluate categorical counts is:

$$\chi^2 = \sum [(O_i - E_i)^2 / E_i]$$

Where O_i represents the observed frequency counts from the sample, and E_i represents the expected frequencies calculated under the null hypothesis.

5.2 Goodness-of-Fit Testing

This framework tests whether an observed sample distribution fits a hypothesized theoretical distribution (such as a uniform, binomial, or normal distribution). The expected frequencies are calculated by multiplying the total sample size by the theoretical probability of each category ($E_i = n * P_i$). The degrees of freedom are defined as:

$$df = k - 1 - m$$

Where k is the number of distinct categories, and m is the number of population parameters estimated from the sample data to build the model.

5.3 Independence Testing via Contingency Tables

This method evaluates whether two categorical variables are independent of each other. The data is organized into an $r \times c$ contingency table, where r represents the rows and c represents the columns. Under the null hypothesis of independence, the expected frequency for each cell is calculated as:

$$E_{ij} = (\text{Row } i \text{ Total} * \text{Column } j \text{ Total}) / \text{Grand Total } N$$

The degrees of freedom for the test are calculated as:

$$df = (r - 1) * (c - 1)$$

6. Analysis of Variance (ANOVA) and Sum of Squares Decompositions

6.1 Theoretical Foundations and Identity Proofs

Analysis of Variance (ANOVA) evaluates whether there are significant differences among the means of three or more independent groups. It works by partitioning the total variation in the data into distinct, measurable components:

$$\text{Total Variation (SST)} = \text{Variation Between Groups (SSB)} + \text{Variation Within Groups (SSW)}$$

The mathematical components for a One-Way ANOVA model are defined below:

- **Grand Total and Correction Factor:** Let T represent the sum of all observations across all groups, and N represent the total number of observations. The Correction Factor (CF) is:

$$CF = T^2 / N$$

- **Total Sum of Squares (SST):** Measures the total variation across all individual data points around the grand mean.

$$SST = \sum \sum x_{ij}^2 - CF$$

- **Sum of Squares Between Groups (SSB):** Tracks the variation between group means, reflecting the effect of the treatment factor.

$$SSB = \sum (T_i^2 / n_i) - CF$$

- **Sum of Squares Within Groups / Error (SSW):** Tracks the natural variation within individual groups, which represents random experimental error.

$$SSW = SST - SSB$$

6.2 The F-Distribution and Variance Ratio Validations

To test the null hypothesis that all group means are equal ($H_0: \mu_1 = \mu_2 = \dots = \mu_k$), we compare the variance between groups to the variance within groups using Mean Squares:

$$MSB = SSB / (k - 1)$$

$$MSW = SSW / (N - k)$$

The final test statistic is the F-Ratio:

$$F = MSB / MSW$$

This calculated value is evaluated against the critical value from an F-distribution table with $df_1 = k - 1$ and $df_2 = N - k$ degrees of freedom at a chosen significance level α .

7. Statistical Quality Control & Three-Sigma Control Systems

7.1 Foundations of Process Stability

Statistical Quality Control (SQC) uses graphical tracking systems called control charts to distinguish between two sources of process variation:

- **Common (Chance) Causes:** Inherently part of the process design, these causes are stable, predictable, and represent random operational noise.
- **Special (Assignable) Causes:** Triggered by specific external events, such as machine malfunctions, raw material changes, or operator errors. These causes create instability and must be quickly identified and corrected.

Control charts use 3-Sigma limits to balance process monitoring efficiency:

$$\text{Control Limits} = \text{Process Baseline Average} \pm 3 * \text{Standard Error}$$

Setting the limits at 3 standard errors ensures that 99.73% of normal variation falls within bounds, minimizing false alarms while reliably catching true process shifts.

7.2 Variable Control Chart Systems (X-bar and R Charts)

Variable charts track continuous physical dimensions by analyzing small subgroups of data collected over regular intervals:

- **X-bar Chart (Process Centering):** Monitors shifts in the process average.

$$CL = \bar{\bar{X}}, \quad UCL = \bar{\bar{X}} + A_2 * \bar{R}, \quad LCL = \bar{\bar{X}} - A_2 * \bar{R}$$

- **R Chart (Process Dispersion):** Monitors changes in process variability by tracking the range within subgroups.

$$CL = \bar{R}, \quad UCL = D_4 * \bar{R}, \quad LCL = D_3 * \bar{R}$$

7.3 Attribute Control Chart Systems (p and c Charts)

Attribute charts track categorical quality metrics, such as counts of defects or proportions of defective units:

- **p-Chart (Proportion Defective):** Tracks the ratio of defective items in samples of size n .

$$UCL / LCL = \bar{p} \pm 3 * \sqrt{[\bar{p} * (1 - \bar{p}) / n]}$$

- **c-Chart (Defect Count):** Tracks the absolute number of individual defects found within a constant unit of production.

$$UCL = \bar{c} + 3 * \sqrt{\bar{c}}, \quad LCL = \bar{c} - 3 * \sqrt{\bar{c}}$$

8. High-Density Mathematical Problem Sets & Multi-Domain Solutions

PROBLEM 1: COMPLEX STRATIFIED RANDOM SAMPLING AND OPTIMUM ALLOCATION DERIVATION

A multinational corporation has a workforce divided into three distinct operational divisions: Engineering ($N_1 = 500$), Operations ($N_2 = 300$), and Administration ($N_3 = 200$). The baseline standard deviations for weekly overtime hours within each division are known to be $\sigma_1 = 8.0$ hours, $\sigma_2 = 4.0$ hours, and $\sigma_3 = 2.0$ hours, respectively. The HR department budgets for a total sample size of $n = 100$ employees. Compute the precise sample sizes required for each stratum using: (a) Proportional Allocation, and (b) Neyman Optimum Allocation.

SOLUTION:

Step 1: Identify given parameters.

$$N_1 = 500, \sigma_1 = 8.0$$

$$N_2 = 300, \sigma_2 = 4.0$$

$$N_3 = 200, \sigma_3 = 2.0$$

$$\text{Total Population } (N) = 500 + 300 + 200 = 1000 \text{ employees}$$

$$\text{Total Targeted Sample size } (n) = 100$$

Step 2: Calculate Part (a) Proportional Allocation.

$$\text{Formula: } n_i = n * (N_i / N)$$

$$\text{Engineering Stratum: } n_1 = 100 * (500 / 1000) = 100 * 0.50 = 50 \text{ employees}$$

$$\text{Operations Stratum: } n_2 = 100 * (300 / 1000) = 100 * 0.30 = 30 \text{ employees}$$

$$\text{Administration Stratum: } n_3 = 100 * (200 / 1000) = 100 * 0.20 = 20 \text{ employees}$$

Verification check: $50 + 30 + 20 = 100$. This distribution matches the population proportions perfectly.

Step 3: Calculate Part (b) Neyman Optimum Allocation.

$$\text{Formula: } n_i = n * (N_i * \sigma_i) / [\sum (N_j * \sigma_j)]$$

First, calculate the product weights for each stratum ($N_i * \sigma_i$):

$$\text{Engineering weight} = 500 * 8.0 = 4000$$

$$\text{Operations weight} = 300 * 4.0 = 1200$$

$$\text{Administration weight} = 200 * 2.0 = 400$$

$$\text{Sum of weights} = 4000 + 1200 + 400 = 5600$$

Now, allocate sample sizes based on these weights:

$$\text{Engineering Sample Size: } n_1 = 100 * (4000 / 5600) = 100 * 0.71428 = 71.43 \approx 71 \text{ employees}$$

$$\text{Operations Sample Size: } n_2 = 100 * (1200 / 5600) = 100 * 0.21428 = 21.43 \approx 21 \text{ employees}$$

$$\text{Administration Sample Size: } n_3 = 100 * (400 / 5600) = 100 * 0.07143 = 7.14 \approx 8 \text{ employees}$$

Verification check: $71 + 21 + 8 = 100$.

Conclusion: Proportional allocation distributes the sample as (50, 30, 20). Neyman optimum allocation adjusts this to (71, 21, 8), shifting more sampling effort to the Engineering division to account for its higher internal variability.

PROBLEM 2: FINITE SAMPLING DISTRIBUTION OF MEANS AND FPC VALIDATION

A cloud computing platform maintains a private server cluster containing exactly $N = 150$ nodes. The network traffic across these nodes has a known mean of 450 GB per day with a standard deviation of 35 GB. A cloud engineer samples 40 nodes at random without replacement. Compute the mean and standard error of this sampling distribution, and calculate the probability that the sample mean traffic exceeds 460 GB.

SOLUTION:

Step 1: Identify given parameters.

Population Size (N) = 150, Sample Size (n) = 40

Population Mean (μ) = 450 GB, Population Standard Deviation (σ) = 35 GB

Sampling Fraction check: $n / N = 40 / 150 = 0.2667$. Since 26.67% of the population is sampled (> 5%), the Finite Population Correction (FPC) factor must be applied.

Step 2: Calculate the parameters of the sampling distribution.

Expected Mean: $\mu_{\bar{x}} = \mu = 450 \text{ GB}$

Standard Error with FPC: $SE(\bar{x}) = (\sigma / \sqrt{n}) * \sqrt{(N - n) / (N - 1)}$

Uncorrected error: $35 / \sqrt{40} = 35 / 6.32455 = 5.5340 \text{ GB}$

FPC factor: $\sqrt{(150 - 40) / (150 - 1)} = \sqrt{110 / 149} = \sqrt{0.73825} = 0.8592$

Corrected Standard Error: $SE(\bar{x}) = 5.5340 * 0.8592 = 4.7548 \text{ GB}$

Step 3: Convert the target boundary to a Standard Normal Z-score.

By the Central Limit Theorem, since $n = 40 \geq 30$, the sampling distribution is approximately normal.

Target Value: $\bar{x} = 460 \text{ GB}$

$Z = (\bar{x} - \mu_{\bar{x}}) / SE(\bar{x}) = (460 - 450) / 4.7548 = 10 / 4.7548 = 2.103 \approx 2.10$

Step 4: Find the cumulative probability from standard normal tables.

We want to find the probability of exceeding this value: $P(Z > 2.10) = 1 - P(Z \leq 2.10)$

From standard Z-tables, the area to the left of $Z = 2.10$ is 0.9821.

$P(Z > 2.10) = 1 - 0.9821 = 0.0179$

Conclusion: The sampling distribution features a mean of 450 GB and a corrected standard error of 4.755 GB. There is a 1.79% probability that the sample mean traffic across the 40 sampled nodes will exceed 46 GB.

PROBLEM 3: ADVANCED CONFIDENCE LIMITS FOR DIFFERENCE OF MEANS (SMALL DEPENDENT/INDEPENDENT)

An automotive research group tests the fuel efficiency of two alternative fuel injector designs, mounting them on small sample engines. Design 1: $n_1 = 12$ engines, sample mean efficiency ($\bar{x}_1$) = 34.2 km/L, sample variance (s_1^2) = 4.5

Design 2: $n_2 = 14$ engines, sample mean efficiency ($\bar{x}_2$) = 31.8 km/L, sample variance (s_2^2) = 3.8

Assuming the fuel efficiency scores follow a normal distribution with equal population variances, construct a 95% confidence interval for the true difference between the mean efficiencies of the two designs ($\mu_1 - \mu_2$).

SOLUTION:

Step 1: Compute the pooled variance estimate.

Given: $n_1 = 12$, $s_1^2 = 4.5$ and $n_2 = 14$, $s_2^2 = 3.8$

Formula: $s_p^2 = [(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2] / [n_1 + n_2 - 2]$

$$s_p^2 = [(12 - 1) * 4.5 + (14 - 1) * 3.8] / [12 + 14 - 2]$$

$$s_p^2 = [11 * 4.5 + 13 * 3.8] / 24 = [49.5 + 49.4] / 24 = 98.9 / 24 = 4.12083$$

Pooled Standard Deviation: $s_p = \sqrt{4.12083} = 2.02998 \text{ km/L}$

Step 2: Determine degrees of freedom and the critical value.

$$df = n_1 + n_2 - 2 = 12 + 14 - 2 = 24$$

For a two-tailed 95% confidence interval with 24 degrees of freedom, the critical value from a Student's t-table is:

$$t_{\alpha/2, 24} = t_{0.025, 24} = 2.064$$

Step 3: Calculate the Standard Error of the Difference.

Formula: $SE = s_p * \sqrt{(1/n_1 + 1/n_2)}$

$$SE = 2.02998 * \sqrt{(1/12 + 1/14)} = 2.02998 * \sqrt{(0.08333 + 0.07143)}$$

$$SE = 2.02998 * \sqrt{0.15476} = 2.02998 * 0.39339 = 0.7986 \text{ km/L}$$

Step 4: Calculate the Margin of Error and Confidence Bounds.

Point Estimate for the difference: $34.2 - 31.8 = 2.4 \text{ km/L}$

$$\text{Margin of Error: } ME = t_{crit} * SE = 2.064 * 0.7986 = 1.6483 \text{ km/L}$$

$$\text{Lower Bound} = 2.4 - 1.6483 = 0.7517 \text{ km/L}$$

$$\text{Upper Bound} = 2.4 + 1.6483 = 4.0483 \text{ km/L}$$

Conclusion: We are 95% confident that the true difference between the mean fuel efficiencies of Design 1 and Design 2 lies between 0.75 km/L and 4.05 km/L. Because this entire range is above zero, we can conclude that Design 1 is significantly more fuel-efficient.

PROBLEM 4: ADVANCED STATISTICAL POWER ANALYSIS AND TYPE II ERROR CALCULATION

A manufacturing plant monitors a chemical packaging process where the target filling weight is $\mu_0 = 50.0$ kg. The filling weights follow a normal distribution with a known population standard deviation of $\sigma = 1.5$ kg. A quality manager tests a random sample of $n = 36$ containers using a significance level of $\alpha = 0.05$ to check if the process is underfilling ($H_1: \mu < 50.0$ kg). If the process actually drifts down to a true mean weight of $\mu_1 = 49.3$ kg, calculate: (a) the probability of committing a Type II error (β), and (b) the statistical power of the test ($1 - \beta$).

SOLUTION:

Step 1: Define the hypothesis framework and find the critical value.

$H_0: \mu \geq 50.0$ kg vs. $H_1: \mu < 50.0$ kg (Left-tailed test)

Significance Level: $\alpha = 0.05$

The critical Z-score for a left-tailed test at $\alpha = 0.05$ is $Z_{crit} = -1.645$.

Step 2: Find the critical boundary value in terms of raw sample means.

Standard Error: $SE = \sigma / \sqrt{n} = 1.5 / \sqrt{36} = 1.5 / 6 = 0.25$ kg

Formula for Critical Value: $\bar{x}_{crit} = \mu_0 + Z_{crit} * SE$

$\bar{x}_{crit} = 50.0 + (-1.645) * 0.25 = 50.0 - 0.41125 = 49.58875$ kg

Decision Rule: We will reject H_0 if any observed sample mean is less than 49.58875 kg.

Step 3: Calculate Part (a) - Probability of a Type II Error (β).

A Type II error occurs if the true mean has shifted to 49.3 kg, but the sample mean still lands above the critical boundary, failing to reject H_0 .

We want to find $P(\bar{x} \geq 49.58875 \text{ given that } \mu = 49.3 \text{ kg})$:

Convert this boundary into a Z-score based on the new true mean ($\mu_1 = 49.3$ kg):

$Z_{new} = (\bar{x}_{crit} - \mu_1) / SE = (49.58875 - 49.3) / 0.25 = 0.28875 / 0.25 = 1.155 \approx 1.16$

Now, find the area to the right of $Z = 1.16$:

From standard Z-tables, the area to the left of 1.16 is 0.8770.

$\beta = P(Z \geq 1.16) = 1 - 0.8770 = 0.1230$

Step 4: Calculate Part (b) - Statistical Power ($1 - \beta$).

Power measures the probability of correctly identifying the process shift and rejecting the false null hypothesis.

$Power = 1 - \beta = 1 - 0.1230 = 0.8770$

Conclusion: If the true process mean drops to 49.3 kg, there is a 12.30% chance the manager will fail to catch the drift (Type II error), and an 87.70% chance (statistical power) that the test will correctly signal the shift and reject the null hypothesis.

PROBLEM 5: COMPREHENSIVE TWO-WAY CONTINGENCY CHI-SQUARE TEST FOR INDEPENDENCE

A software engineering division investigates whether the frequency of post-release system bugs is independent of the primary programming language used to build applications. Over a testing cycle, 500 modules are audited, yielding the following observed frequency distributions:

Programming Language	Low Bugs	Medium Bugs	High Bugs	Row Totals
Java	70	60	20	150
C++	45	65	40	150
Python	95	85	20	200
Column Totals	210	210	80	500 (Grand Total)

Using a 1% significance level, test whether module bug frequency is independent of the programming language used.

SOLUTION:

Step 1: State hypotheses and determine the critical value.

H_0 : Bug frequency is independent of the programming language used.

H_1 : Bug frequency is dependent on the programming language used.

Calculate Degrees of Freedom: $df = (rows - 1) * (columns - 1) = (3 - 1) * (3 - 1) = 2 * 2 = 4$

From a Chi-Square distribution table, the critical value for $df = 4$ at $\alpha = 0.01$ is $\chi^2_{crit} = 13.277$.

Step 2: Calculate Expected Frequencies ($E = \text{Row Total} * \text{Column Total} / \text{Grand Total}$).

- For Java (Row Total = 150):
Expected Low = $(150 * 210) / 500 = 63$
Expected Medium = $(150 * 210) / 500 = 63$
Expected High = $(150 * 80) / 500 = 24$
- For C++ (Row Total = 150):
Expected Low = $(150 * 210) / 500 = 63$
Expected Medium = $(150 * 210) / 500 = 63$
Expected High = $(150 * 80) / 500 = 24$
- For Python (Row Total = 200):
Expected Low = $(200 * 210) / 500 = 84$

$$\text{Expected Medium} = (200 * 210) / 500 = 84$$

$$\text{Expected High} = (200 * 80) / 500 = 32$$

Step 3: Compute the Chi-Square statistic using the formula $\sum [(O - E)^2 / E]$.

- Java terms:

$$\text{Low: } (70 - 63)^2 / 63 = 49 / 63 = 0.7778$$

$$\text{Medium: } (60 - 63)^2 / 63 = 9 / 63 = 0.1429$$

$$\text{High: } (20 - 24)^2 / 24 = 16 / 24 = 0.6667$$

- C++ terms:

$$\text{Low: } (45 - 63)^2 / 63 = (-18)^2 / 63 = 324 / 63 = 5.1429$$

$$\text{Medium: } (65 - 63)^2 / 63 = 4 / 63 = 0.0635$$

$$\text{High: } (40 - 24)^2 / 24 = 16^2 / 24 = 256 / 24 = 10.6667$$

- Python terms:

$$\text{Low: } (95 - 84)^2 / 84 = 11^2 / 84 = 121 / 84 = 1.4405$$

$$\text{Medium: } (85 - 84)^2 / 84 = 1 / 84 = 0.0119$$

$$\text{High: } (20 - 32)^2 / 32 = (-12)^2 / 32 = 144 / 32 = 4.5000$$

Summing all terms: $\chi^2_{calc} = 0.7778 + 0.1429 + 0.6667 + 5.1429 + 0.0635 + 10.6667 + 1.4405 + 0.0119 + 4.5000 = 23.4129$

Step 4: Compare against the critical boundary and make a decision.

Our calculated value is $\chi^2_{calc} = 23.413$, and the critical value is $\chi^2_{crit} = 13.277$. Because $23.413 > 13.277$, the statistic falls deep within the rejection region.

Conclusion: We reject the null hypothesis at the 1% significance level. There is strong statistical evidence that post-release bug frequencies depend significantly on the choice of programming language.

PROBLEM 6: COMPREHENSIVE ONE-WAY ANOVA MATRIX AND F-TEST VALIDATION

An enterprise organization tests the query execution performance of four database engine configurations. The response times (in milliseconds) from five identical test queries are recorded below:

- Config 1: 14, 16, 12, 18, 15 (Total $T_1 = 75$, sample size $n_1 = 5$)
- Config 2: 22, 26, 20, 24, 23 (Total $T_2 = 115$, sample size $n_2 = 5$)
- Config 3: 10, 12, 9, 14, 10 (Total $T_3 = 55$, sample size $n_3 = 5$)
- Config 4: 18, 20, 16, 22, 19 (Total $T_4 = 95$, sample size $n_4 = 5$)

Using a 5% significance level, perform a One-Way ANOVA to test whether there are significant differences in the mean execution times across the configurations.

SOLUTION:

Step 1: State hypotheses and summarize basic metrics.

$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$ (All configuration means are equal)

H_1 : At least one configuration mean is significantly different.

Number of groups (k) = 4, Sample size per group (n) = 5, Total observations (N) = 20

Grand Total (T) = $75 + 115 + 55 + 95 = 340$

Correction Factor (CF) = $T^2 / N = 340^2 / 20 = 115600 / 20 = 5780$

Step 2: Calculate individual squared raw metrics ($\sum \Sigma x^2$).

Config 1: $14^2 + 16^2 + 12^2 + 18^2 + 15^2 = 196 + 256 + 144 + 324 + 225 = 1145$

Config 2: $22^2 + 26^2 + 20^2 + 24^2 + 23^2 = 484 + 676 + 400 + 576 + 529 = 2665$

Config 3: $10^2 + 12^2 + 9^2 + 14^2 + 10^2 = 100 + 144 + 81 + 196 + 100 = 621$

Config 4: $18^2 + 20^2 + 16^2 + 22^2 + 19^2 = 324 + 400 + 256 + 484 + 361 = 1825$

Sum of all squared observations = $1145 + 2665 + 621 + 1825 = 6256$

Step 3: Partition the Sum of Squares.

Total Sum of Squares: $SST = \sum \Sigma x^2 - CF = 6256 - 5780 = 476$

Sum of Squares Between Groups: $SSB = [\sum T_i^2 / n_i] - CF$

$SSB = [(75^2 / 5) + (115^2 / 5) + (55^2 / 5) + (95^2 / 5)] - 5780$

$SSB = [(5625 / 5) + (13225 / 5) + (3025 / 5) + (9025 / 5)] - 5780$

$SSB = [1125 + 2645 + 605 + 1805] - 5780 = 6180 - 5780 = 400$

Sum of Squares Within Groups (Error): $SSW = SST - SSB = 476 - 400 = 76$

Step 4: Compute Mean Squares and the final F-Ratio.

Degrees of Freedom: Between Groups = $k - 1 = 3$; Within Groups = $N - k = 20 - 4 = 16$

Mean Square Between: $MSB = SSB / 3 = 400 / 3 = 133.333$

Mean Square Within: $MSW = SSW / 16 = 76 / 16 = 4.750$

Computed F-Ratio: $F_{calc} = MSB / MSW = 133.333 / 4.750 = 28.07$

Step 5: Compare against the critical boundary.

From an F-distribution table, the critical value for $df = (3, 16)$ at $\alpha = 0.05$ is $F_{crit} = 3.24$. Because $F_{calc} = 28.07 > 3.24$, the null hypothesis is rejected.

Complete ANOVA Summary Table:

Source of Variation	Sum of Squares (SS)	Degrees of Freedom (df)	Mean Square (MS)	Computed F-Ratio
Between Configurations	400.00	3	133.333	28.07
Within Configurations (Error)	76.00	16	4.750	-
Total Baseline	476.00	19	-	-

Conclusion: We reject the null hypothesis at the 5% significance level. There is a highly significant difference in the mean query execution times across the four database engine configurations.

PROBLEM 7: ADVANCED ATTRIBUTE CONTROL SYSTEMS VIA FRACTION DEFECTIVE P-CHARTS

A manufacturing electronics assembly line monitors quality by inspecting samples of $n = 100$ printed circuit boards every hour. After tracking production for 25 consecutive hours, inspectors count a total of 150 defective boards. Calculate the process baseline defective proportion, and compute the Center Line (CL), Upper Control Limit (UCL), and Lower Control Limit (LCL) for a standard 3-Sigma fraction defective p-chart.

SOLUTION:

Step 1: Identify given baseline parameters.

Total subgroups examined (k) = 25

Sample size per subgroup (n) = 100 boards

Total defective items found across all samples = 150 boards

Total items inspected overall = $k * n = 25 * 100 = 2500$ boards

Step 2: Calculate the baseline average defective proportion ($p\text{-bar}$).

$$p\text{-bar} = \text{Total Defectives} / \text{Total Inspected} = 150 / 2500 = 0.06$$

Step 3: Compute the 3-Sigma Control Limits.

Center Line (CL) = $p\text{-bar} = 0.06$

Standard Error of Proportions: $SE(p) = \sqrt{[p\text{-bar} * (1 - p\text{-bar}) / n]}$

$$SE(p) = \sqrt{[0.06 * (1 - 0.06) / 100]} = \sqrt{[0.06 * 0.94 / 100]} = \sqrt{[0.0564 / 100]} = \sqrt{0.000564} = 0.023748$$

Calculate limits using 3 standard errors:

Upper Control Limit: $UCL = p\text{-bar} + 3 * SE(p) = 0.06 + 3 * 0.023748 = 0.06 + 0.07124 = 0.13124$

Lower Control Limit: $LCL = p\text{-bar} - 3 * SE(p) = 0.06 - 3 * 0.023748 = 0.06 - 0.07124 = -0.01124$

Since a defective proportion cannot be negative, the Lower Control Limit is adjusted to exactly 0.00.

Summary Table of Control Boundaries:

Chart Boundary Metric	Calculated Value	Final Applied Limit
Lower Control Limit (LCL)	-0.01124	0.0000 (0.0%)
Center Line (CL)	0.06000	0.0600 (6.0%)
Upper Control Limit (UCL)	0.13124	0.1312 (13.12%)

Conclusion: The baseline process defective rate is 6.0%. The control limits are set at an LCL of 0.0% and an UCL of 13.12%. If any future hourly sample reveals a defective rate exceeding 13.12%, it signals that the process has gone out of control due to an assignable cause.